Business Intelligence Analytics

The end of the year 2015, a Pizza business is looking to anayse the performance of the business over the year. They are interested in tracking KPI performance, trends over time and visualization that shows these trends.

My Process

Overview

This is the First Part of a two part Sales analysis.

This part involved using Excel and SQL to clean and analyse data provided by a client that runs a Pizza business.

The data information was for the entire year of 2015 and contained 48620 rows worth of data.

Along the way hurdles were encoutered in the analysis of this data but with proper research and knowledge they were quickly cleared and the problem statements were effectively tackled

Although Problem statements from the secont Part were also tackled here using SQL, the charts will be better visualized using Power BI

Business Problem Statement(EDA)

KPI REQUIREMENT

1. **TOTAL REVENUE:** The sum of total price of all pizza orders
2. **AVERAGE ORDER VALUE:** The average anount spent per order, calsulated by dividing the total revenue by the total number of orders
3. **TOTAL PIZZAS SOLD:** The sum of the quantities of all pizza sold
4. **TOTAL ORDERS:** The total number of orders placed.
5. **AVG PIZZAS PER ORDER:** The avg number of pizzas sold per order, calculated by dividing the total number of pizzas sold by the total number of orders

Business Problem Statement(EDA)

CHARTS REQUIREMENT

6. Daily Trend for Total Orders

7. Monthly Trend for total Orders

8. Percentage of Sales by Pizza Category

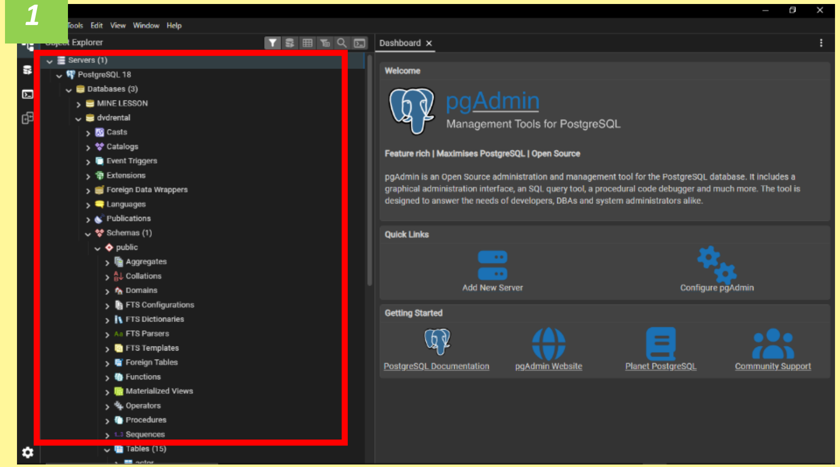
9. Percentage of Sales by Pizza size

10. Top 5 Best sellers by Revenue, Total Quantiry and total orders

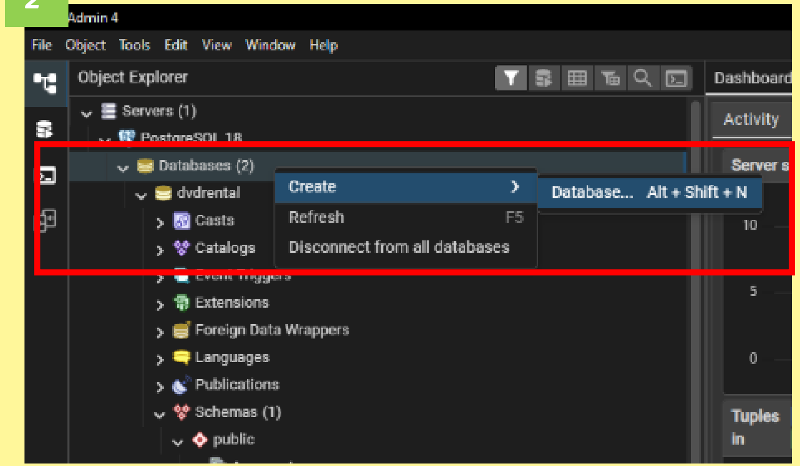
11. Bottom 5 worst sellers by Revenue, Total Quantiry and total orders

1. Creating DataBase

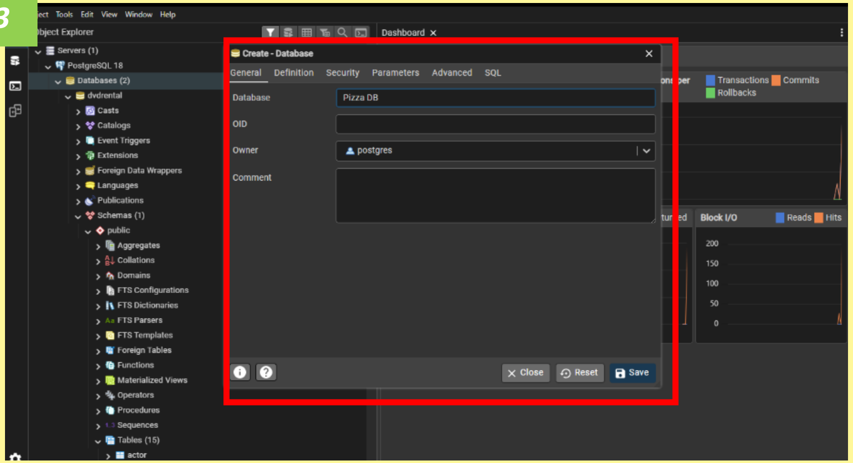
1



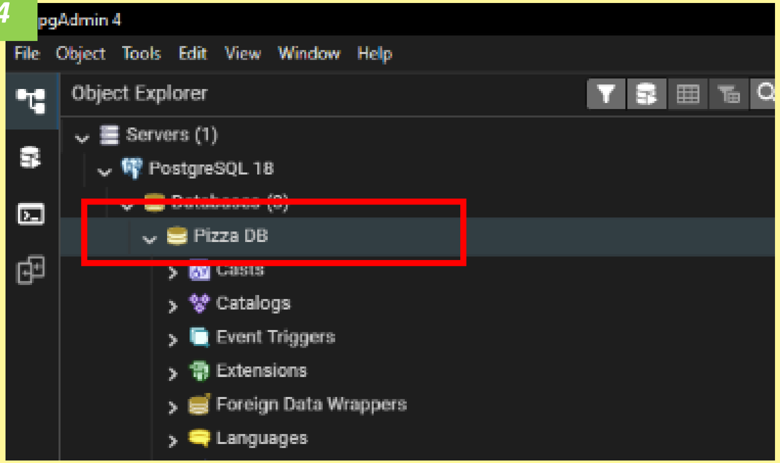
2



3



4



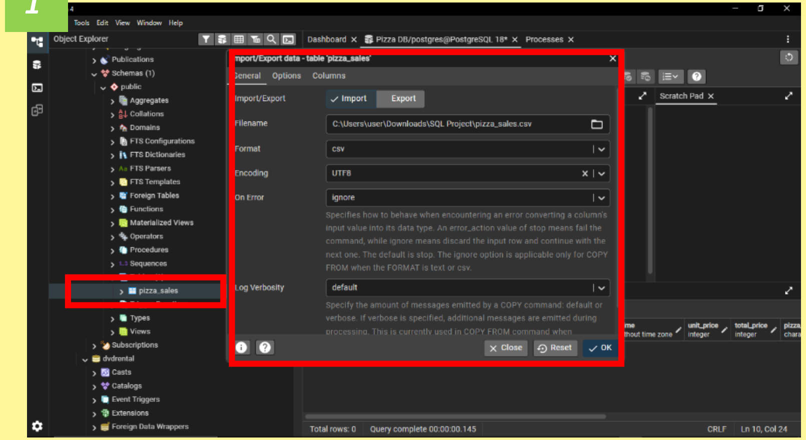
2. Creating Table and importing csv file

```sql
Create table pizza_sales(
pizza_id int primary key,
order_id int,
pizza_name_id varchar (50),
quantity int,
order_date date,
order_time time,
unit_price int,
total_price int,
pizza_size varchar (5),
pizza_category varchar (50),
pizza_ingredients varchar (500),
pizza_name varchar (50)
);
select *from
```

SQL to create a table, note all columes must match with that of the intended data to import

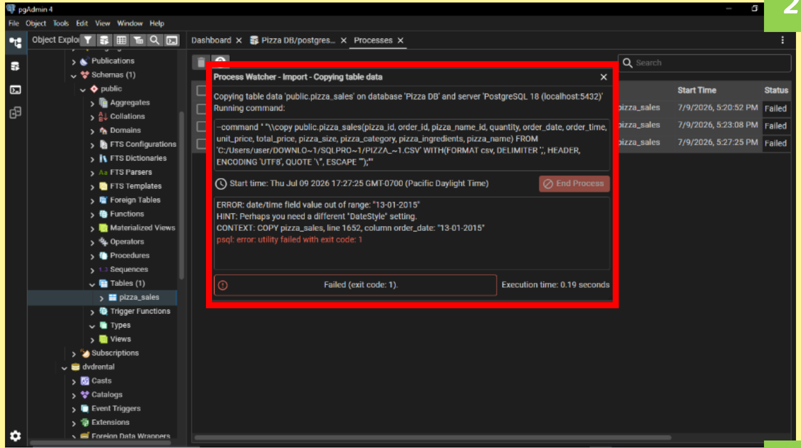
3. Importing File, Data Cleaning and resolving errors with importing csv file

1



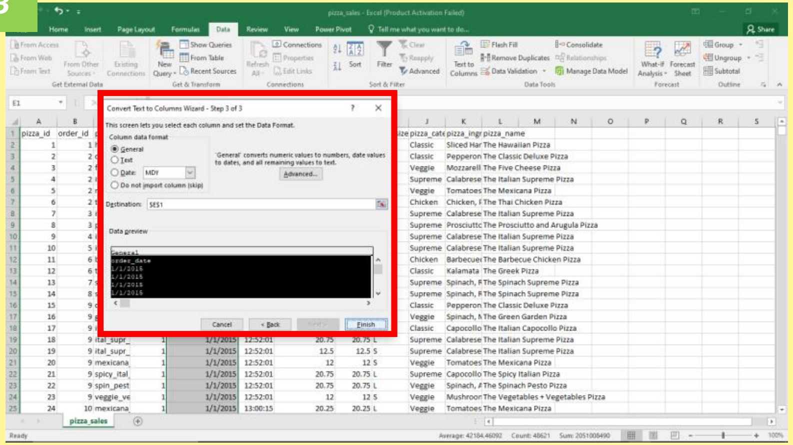
1st attempt to import csv file

2



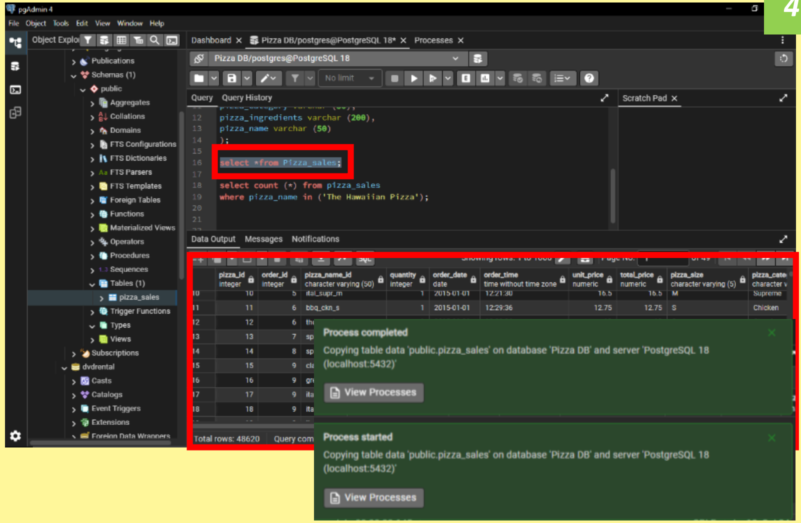
Failed to import. Error message

3

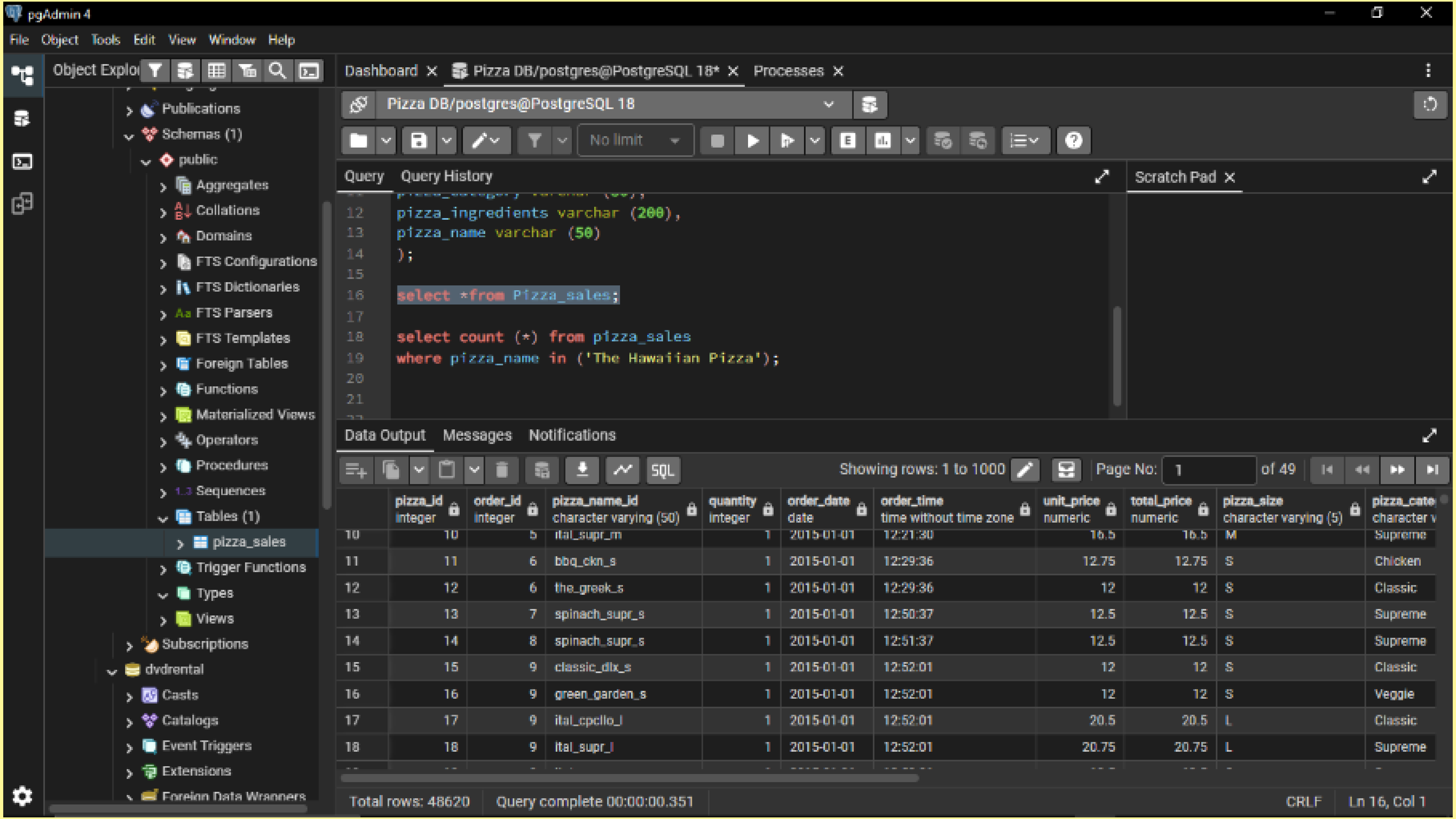


Cleaned file with Excel

4



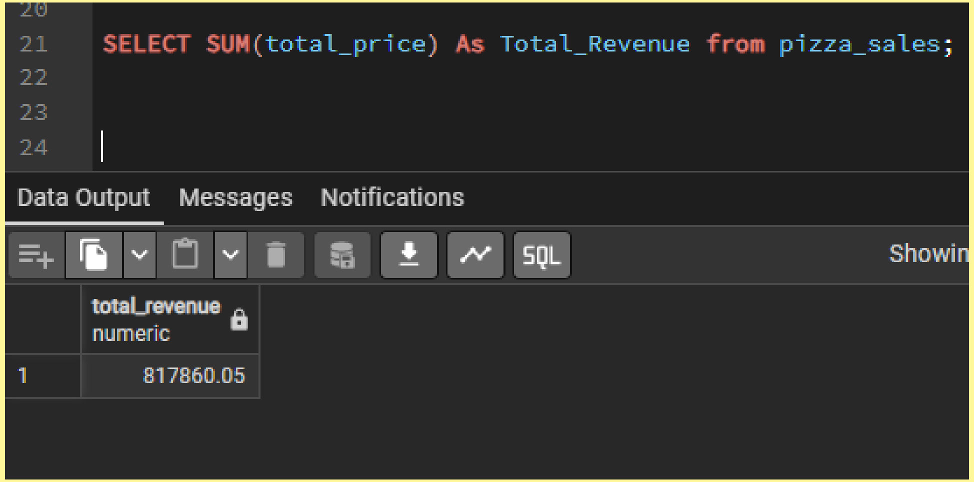
2nd attempt was successful with a confirmation pop up


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Dashboard
Pizza DB/postgres@PostgreSQL 18*
Processes
Pizza DB/postgres@PostgreSQL 18
No limit
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (1)
pizza_sales
Trigger Functions
Types
Views
Subscriptions
dvdrental
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Query
Query History
Scratch Pad
pizza_ingredients varchar (200),
pizza_name varchar (50)
);
select *from Pizza_sales;
select count (*) from pizza_sales
where pizza_name in ('The Hawaiian Pizza');
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1000
Page No: 1 of 49
pizza_id integer | order_id integer | pizza_name_id character varying (50) | quantity integer | order_date date | order_time time without time zone | unit_price numeric | total_price numeric | pizza_size character varying (5) | pizza_category character v
11 | 11 | 6 | bbq_ckn_s | 1 | 2015-01-01 | 12:29:36 | 12.75 | 12.75 | S | Chicken
12 | 12 | 6 | the_greek_s | 1 | 2015-01-01 | 12:29:36 | 12 | 12 | S | Classic
13 | 13 | 7 | spinach_supr_s | 1 | 2015-01-01 | 12:50:37 | 12.5 | 12.5 | S | Supreme
14 | 14 | 8 | spinach_supr_s | 1 | 2015-01-01 | 12:51:37 | 12.5 | 12.5 | S | Supreme
15 | 15 | 9 | classic_dlx_s | 1 | 2015-01-01 | 12:52:01 | 12 | 12 | S | Classic
16 | 16 | 9 | green_garden_s | 1 | 2015-01-01 | 12:52:01 | 12 | 12 | S | Veggie
17 | 17 | 9 | ital_cpcllo_l | 1 | 2015-01-01 | 12:52:01 | 20.5 | 20.5 | L | Classic
18 | 18 | 9 | ital_supr_l | 1 | 2015-01-01 | 12:52:01 | 20.75 | 20.75 | L | Supreme
Total rows: 48620
Query complete 00:00:00.351
CRLF
Ln 16, Col 1

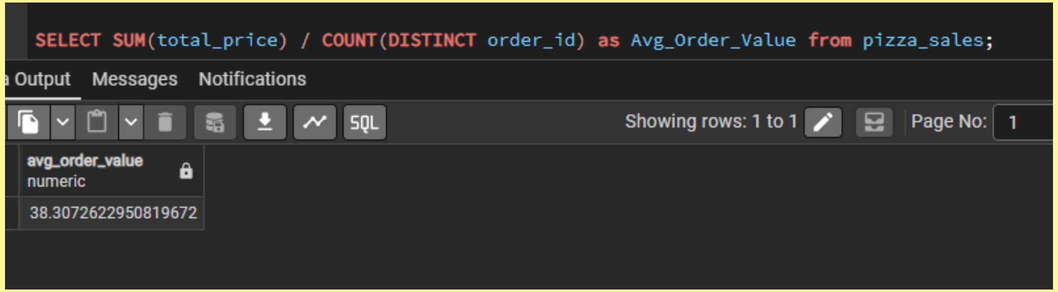
ANSWERING BUSINESS PROBLEM STATEMENT

After cleaning and importing the data, we move to the querying proper

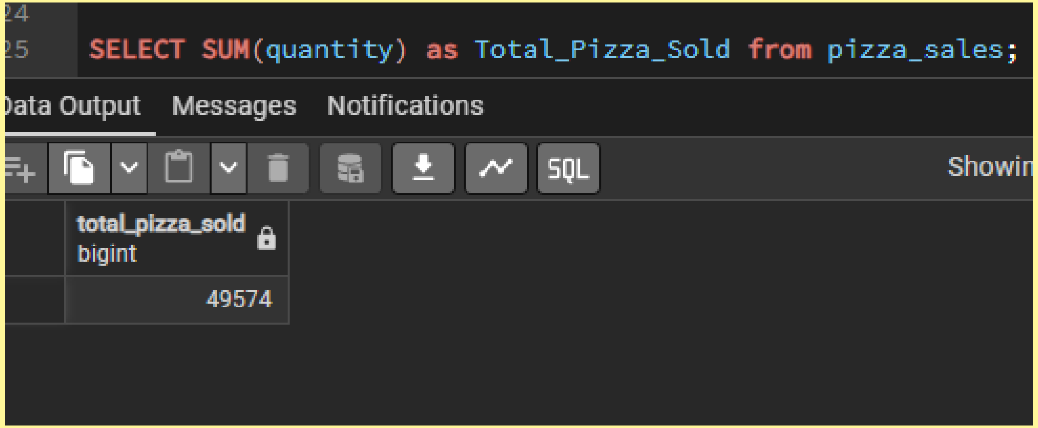
1. TOTAL REVENUE



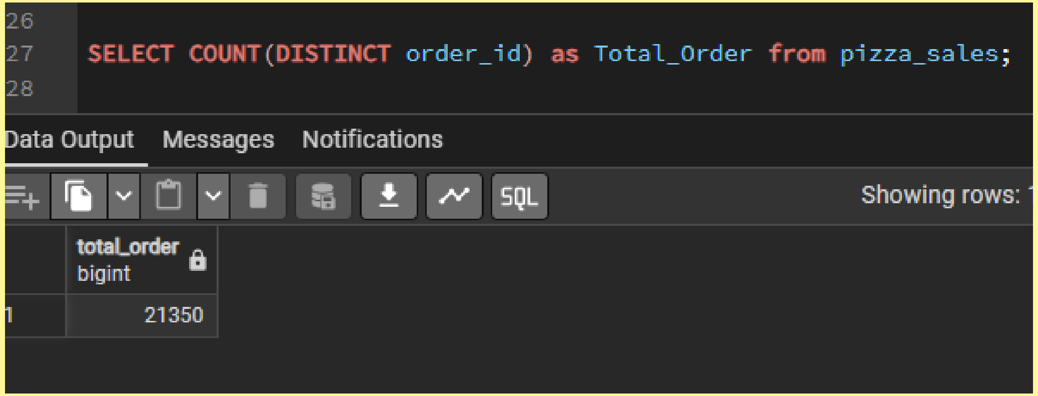
2. AVERAGE ORDER VALUE



3. TOTAL PIZZAS SOLD



4. TOTAL ORDERS



5. AVG PIZZAS PER ORDER

```sql
SELECT CAST (CAST (SUM(quantity) AS DECIMAL (10,2)) /
CAST (COUNT(DISTINCT order_id) AS DECIMAL (10,2)) AS DECIMAL (10,2))
AS Avg_Pizza_Per_Order FROM pizza_sales;
```

ata Output Messages Notifications

SQL Showing rows: 1 to 1

avg_pizza_per_order numeric (10,2)
2.32

6. Daily Trend for Total Orders

```sql
SELECT
    TO_CHAR(order_date, 'Dy') AS order_day,
    COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY TO_CHAR(order_date, 'Dy')
ORDER BY MIN(EXTRACT(DOW FROM order_date));
```

Data Output Messages Notifications

	order_day text	total_orders bigint
1	Sun	2624
2	Mon	2794
3	Tue	2973
4	Wed	3024
5	Thu	3239
6	Fri	3538
7	Sat	3158

7. Monthly Trend for total Orders

```sql
SELECT
    TO_CHAR(order_date, 'Mon') AS order_month,
    COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY TO_CHAR(order_date, 'Mon'), EXTRACT(MONTH FROM order_date)
ORDER BY EXTRACT(MONTH FROM order_date);
```

Data Output Messages Notifications

Showing rows: 1 to 1

	order_month text	total_orders bigint
1	Jan	1845
2	Feb	1685
3	Mar	1840
4	Apr	1799
5	May	1853
6	Jun	1773
7	Jul	1935
8	Aug	1841
9	Sep	1661
10	Oct	1646
11	Nov	1792
12	Dec	1680

Total rows: 12 Query complete 00:00:00.497

8. Percentage of Sales by Pizza Category

```sql
SELECT pizza_category, CAST(SUM(total_price) * 100.0 /
(SELECT SUM(total_price) FROM pizza_sales)
AS DECIMAL(10,2)) AS percentage_of_sales
FROM pizza_sales
GROUP BY pizza_category;
```

Data Output Messages Notifications

	pizza_category character varying (50)	percentage_of_sales numeric (10,2)
1	Supreme	25.46
2	Chicken	23.96
3	Veggie	23.68
4	Classic	26.91

9. Percentage of Sales by Pizza size

```sql
SELECT pizza_size, CAST(SUM(total_price) * 100.0 /
(SELECT SUM(total_price) FROM pizza_sales)
AS DECIMAL(10,2)) AS percentage_of_sales
FROM pizza_sales
GROUP BY pizza_size;
```

Data Output Messages Notifications

	pizza_size character varying (5)	percentage_of_sales numeric (10,2)
1	S	21.77
2	XXL	0.12
3	XL	1.72
4	M	30.49
5	L	45.89

10. Total Pizzas sold by pizza category

```sql
select pizza_category, count(distinct order_id)
as Total_orders from pizza_sales
group by pizza_category
order by Total_orders desc
limit 5
```

ta Output Messages Notifications

pizza_category character varying (50)	total_orders bigint
Classic	10859
Supreme	9085
Veggie	8941
Chicken	8536

11. Top 5 Best sellers by Revenue, Total Quantity and total orders

```sql
SELECT pizza_name, SUM(Quantity) Total_Quantity from pizza_sales
group by pizza_name
Order by Total_Quantity DESC
LIMIT 5
```

Data Output Messages Notifications

Showing rows: 1

	pizza_name character varying (50)	total_quantity bigint
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Piz...	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

Total rows: 5 Query complete 00:00:00.139

```sql
SELECT pizza_name, SUM(total_price) Total_Revenue from pizza_sales
group by pizza_name
Order by Total_Revenue DESC
LIMIT 5
```

Data Output Messages Notifications

Showing rows: 1 to 5

	pizza_name character varying (50)	total_revenue numeric
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Piz...	42768.00
3	The California Chicken Piz...	41409.50
4	The Classic Deluxe Pizza	38180.5
5	The Spicy Italian Pizza	34831.25

```sql
SELECT pizza_name, COUNT(DISTINCT order_id) AS Total_orders from pizza_sales
group by pizza_name
Order by Total_orders DESC
LIMIT 5
```

Data Output Messages Notifications

Showing rows: 1 to 5

	pizza_name character varying (50)	total_orders bigint
1	The Classic Deluxe Pizza	2329
2	The Hawaiian Pizza	2280
3	The Pepperoni Pizza	2278
4	The Barbecue Chicken Piz...	2273
5	The Thai Chicken Pizza	2225

12. Bottom 5 worst sellers by Revenue, Total Quantiry and total orders

```sql
SELECT pizza_name, SUM(total_price) Total_Revenue from pizza_sales
group by pizza_name
Order by Total_Revenue ASC
LIMIT 5
```

Data Output | Messages | Notifications

Showing rows: 1 to

	pizza_name character varying (50)	total_revenue numeric
1	The Brie Carre Pizza	11588.50
2	The Green Garden Pizza	13955.75
3	The Spinach Supreme Piz...	15277.75
4	The Mediterranean Pizza	15360.50
5	The Spinach Pesto Pizza	15596.00

```sql
SELECT pizza_name, SUM(Quantity) Total_Quantity from pizza_sales
group by pizza_name
Order by Total_Quantity ASC
LIMIT 5
```

Data Output | Messages | Notifications

Showing rows: 1

	pizza_name character varying (50)	total_quantity bigint
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Piz...	950
5	The Soppressata Pizza	961

Total rows: 5 Query complete 00:00:00.126

```sql
SELECT pizza_name, COUNT(DISTINCT order_id) AS Total_orders from pizza_sales
group by pizza_name
Order by Total_orders
LIMIT 5
```

Data Output | Messages | Notifications

Showing rows: 1 to 5

	pizza_name character varying (50)	total_orders bigint
1	The Brie Carre Pizza	480
2	The Mediterranean Pizza	912
3	The Calabrese Pizza	918
4	The Spinach Supreme Piz...	918
5	The Chicken Pesto Pizza	938

Conclusion

- In the cource of this project I faced a number of blockers, most of which were at the initial stage and affected the total number of rows imported into PostgreSQL and also in some cases stopped the importaation with an error message.
- With my good research skills and with Excel I was able to locate and clean the data allowin gme to move forward.
- Queries like WHERE, GROUP BY, ORDER BY, CASE were particularly handy in this project.
- Although the JOIN statement wasn’t used here it remains another very powerful tool in SQL.

Thank you for your time.
I would be more than happy to discuss further and address any concerns you may have.

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